

Synopsis

TITLE	Hybrid Post-Quantum VPN for Secure Remote Access	
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1. Introduction

Virtual Private Networks protect data over untrusted networks, but most deployed VPN handshakes still rely on classical public-key cryptography that can be broken by future large-scale quantum computers. This project proposes a compact hybrid VPN prototype for secure remote access, where session keys are derived from both X25519 elliptic-curve Diffie–Hellman and the post-quantum ML-KEM-768 key encapsulation mechanism. The objective is to demonstrate a practical migration path that preserves present-day performance while reducing the “harvest now, decrypt later” risk for long-lived confidential traffic.

2. Existing System

Conventional VPN solutions such as WireGuard, OpenVPN and IPsec primarily use ECC- or RSA-based authentication and key exchange. These systems provide strong security against classical attackers, but their public-key layer is vulnerable to quantum attacks based on Shor’s algorithm. Existing deployments also treat post-quantum protection as an add-on rather than an integrated design goal, which creates uncertainty in long-term confidentiality, downgrade resistance and algorithm agility during future migration.

3. Proposed System

The proposed system is a hybrid VPN tunnel with a Python-based backend and a Next.js graphical user interface. By combining classical and post-quantum key establishment in a single handshake, it ensures session keys remain secure against future cryptographic weaknesses.

Key operations include:

- **Key Exchange:** Parallel execution of X25519 (classical) and ML-KEM-768 (post-quantum).
- **Key Derivation:** Both secrets are combined using HKDF-SHA256 for final session keys.
- **Encryption:** Tunnel traffic is secured using hardware-accelerated AES-GCM.
- **Authentication:** A hybrid signature model uses ECDSA-P256 and ML-DSA to avoid reliance on a single scheme.

The design also includes secure algorithm negotiation, packet-level integrity verification, and periodic rekeying. The system will be evaluated for secure file transfer and remote access

against classical-only VPNs to demonstrate practical hybrid cryptography migration.

4. Technology Usage

The project combines modern web technologies and cryptographic libraries. The backend uses Python for rapid prototyping and robust security, while the client features a modern Next.js and Tailwind CSS graphical user interface.

Major technologies include:

Category	Technologies
Frontend	Next.js, React, Tailwind CSS, TypeScript, Axios
Backend	Python, FastAPI, AsyncIO, Cryptography
Classical Cryptography	OpenSSL (X25519, ECDSA-P256, AES-GCM)
Post-Quantum Cryptography	liboqs (ML-KEM-768, ML-DSA)
Networking & System	Linux TUN/TAP, Sockets

This stack offers a realistic platform for evaluating hybrid quantum-safe communications against conventional VPN setups.

5. Societal Concern

The project addresses the real-world need to protect sensitive communication that must remain confidential for many years, including healthcare records, academic research, enterprise intellectual property and government data. By demonstrating a feasible hybrid VPN design, the work supports early adoption of quantum-safe networking practices and helps institutions transition to stronger security without abandoning currently trusted systems.

6. Possible Outcome

The expected outcome is a working proof-of-concept hybrid VPN prototype with experimental results that compare classical and hybrid handshakes in terms of security and performance. The project can lead to a technical report or conference-style paper on post-quantum VPN migration, and the implementation may serve as a baseline for future extension into WireGuard-, TLS- or IPsec-based deployments.

Signature of Faculty